An Attempt At Encoding March Madness

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Abstract — Here I will present a weighted-vector approach to tournament simulation, specifically addressing the volatility of the NCAA Division I Men's Basketball Tournament. By utilizing a user-defined objective function, the model allows for the prioritization of specific basketball philosophies—ranging from defensive stalwarts to high-variance "chaos" factors—to generate a non-static probability space for matchup outcomes.

I. User-Defined Objective Function

The strength of a team, T, is calculated using a linear combination of normalized performance metrics, where the coefficients ω are provided via user input:

$$S(T) = \omega_{off} \cdot \text{Offense}_T + \omega_{def} \cdot \text{Defense}_T + \omega_{chaos} \cdot \text{Chaos}_T$$

To maintain a consistent probability space across various strategies, the model enforces the following stochastic constraint:

$$\sum \omega = 1.0$$

II. Matchup Probability and Seed-Bias

To determine the outcome of a matchup between T_1 and T_2, the model generates a base win probability (P_{base}) derived from the scores. A *Seed Boost* is then added to reflect the historical advantage of high-seeded teams:

$$P(T_1) = \text{clamp}\left(\frac{S(T_1)}{S(T_1) + S(T_2)} + 0.025(Seed_2 - Seed_1)\right)$$

The result is clamped to the range $[0.05, 0.95]$, preserving the ”Madness” potential for extreme upsets (e.g., a 16-seed over a 1-seed).

III. Monte Carlo Simulation

The winner is determined by a single-trial Monte Carlo method. For each matchup, a random variable $X \sim U(0, 1)$ is sampled. If $X \leq P(T_1)$, Team 1 advances; otherwise, we have an upset, and Team 2 is declared the winner.